



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(Accredited by NBA Tier 1: 2025-2028)

Project Phase- I Report on
[22CS76]
A Web-Based Multimodal AI System for Early Detection of Dyslexia

Submitted in partial fulfillment for the award of the Degree of

Bachelor of Engineering
in
Computer Science and Engineering

Submitted by

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Under the Guidance of

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JNANASANGAMA, BELAGAVI-590018, KARNATAKA, INDIA
2025-26

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CERTIFICATE

Certified that the project report entitled “A Web-Based Multimodal AI System for Early Detection of Dyslexia” carried out by **B Sathvika Reddy (USN: 1DS22CS056)**, **Jyothika Lakshmi M (USN: 1DS22CS097)**, **N Shreya Sai (USN: 1DS22CS132)**, **Vrinda B S (USN: 1DS22CS256)** are bonafide students of **DAYANANDA SAGAR COLLEGE OF ENGINEERING**, an autonomous institution affiliated to VTU, Belagavi in partial fulfillment for the award of Degree of **Bachelor of Computer Science and Engineering** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Project Phase-I report has been approved as it satisfies the academic requirements with respect to the work prescribed for the said Degree.

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DECLARATION

We, **B Sathvika Reddy (1DS22CS056)**, **Jyothika Lakshmi M (1DS22CS097)**, **N Shreya Sai (1DS22CS132)** and **Vrinda B S (1DS22CS256)**, respectively, hereby declare that the project work entitled “ **A Web-Based Multimodal AI System for Early Detection of Dyslexia** ” has been independently done by us under the guidance of ‘**Dr. Ramya R S**’, **Associate Professor**, CSE department and submitted in partial fulfillment of the requirement for the award of the degree of **Bachelor of Computer Science and Engineering** at **Dayananda Sagar College of Engineering**, an autonomous institution affiliated to VTU, Belagavi during the academic year 2025-2026.

We further declare that we have not submitted this report either in part or in full to any other university for the award of any degree.

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ABSTRACT

(a) Objectives and Goals of the Project

The objective of this project is to develop an accessible, web-based early detection system for dyslexia in children. The goal is to identify dyslexia-related patterns using interactive digital assessments and machine learning models, enabling timely intervention and improved learning outcomes.

(b) Basic Methods and Procedures Used

The system incorporates three interactive activities—letter tracing, reading-aloud tasks, and memory-based games—to collect handwriting, speech, and cognitive data. These inputs are analyzed using specific machine learning approaches: CNN models for handwriting stroke classification, BiLSTM networks for speech-sequence analysis, and Random Forest classifiers for cognitive task evaluation. All modules are integrated into a unified web platform for wide accessibility and ease of use.

(c) Main Results and Outcomes

Results show that combining handwriting dynamics, speech sequencing patterns, and memory-task performance yields significantly higher dyslexia prediction accuracy compared to traditional, single-feature assessments. The system generates real-time dyslexia risk scores and provides insights into specific difficulty areas, making the evaluation process more reliable and scalable.

(d) Applications and Future Scope

The proposed system can be applied in schools, pediatric assessment centers, and remote learning environments for large-scale screening of dyslexia. Future enhancements include multilingual speech analysis, adaptive activity difficulty levels, mobile app integration, and expansion toward detecting other learning disabilities. The platform can evolve into a comprehensive, multimodal cognitive screening tool.

Keywords: Dyslexia, Early Screening, Multimodal Data, Handwriting Analysis, Speech-to-Text, Memory Tasks, LSTM, BiLSTM, Deep Learning, Feature Extraction, Web-Based Platform, Flask API, MongoDB

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LIST OF ABBREVIATIONS

Abbreviation	Full Description
AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
LSTM	Long Short Term Memory
BiLSTM	Bi-directional LSTM
STT	Speech To Text
UI	User Interface
API	Application Programming Inteface
JSON	Java Script Object Notation
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Overview

Early identification of learning disabilities is essential for ensuring timely support and improved academic outcomes. Dyslexia, a widely prevalent learning disorder, affects a child's ability to read, write, and process language despite normal intelligence and adequate schooling. Traditional diagnostic procedures require trained professionals, involve manual assessments, and are often inaccessible to many regions, leading to delayed interventions.

To address these challenges, this project proposes a web-based multimodal AI system designed to detect early indicators of dyslexia using child-friendly interactive tasks. The system captures three key behavioral inputs: handwriting patterns through letter-tracing on a digital canvas, reading ability through speech-to-text-based reading tasks, and working memory through short memory-recall games. These multimodal inputs are processed into structured features and analyzed using deep learning models, primarily BiLSTM and LSTM networks, which are well-suited for sequential and temporal data.

The complete solution is implemented using a modular architecture consisting of a browser-based user interface, a MongoDB database for secure data storage, and a trained neural model to classify dyslexia risk. The proposed system aims to provide an accessible, cost-effective, and scalable platform that can support schools, parents, and clinical practitioners in screening children at an early stage. By utilizing interactive tasks and machine learning, the project bridges the gap between traditional assessment methods and modern digital diagnostics, offering a meaningful step toward inclusive and technology-driven educational support.

1.2 Problem Statement

Dyslexia often goes undetected during early schooling years due to the reliance on manual evaluations conducted by trained specialists. These assessments are time-consuming, subjective, and not easily accessible in many regions, leaving children without timely support. Traditional tools also assess only a single skill such as reading or handwriting making early detection inaccurate and incomplete. There is a need for a scalable, automated system that can analyze multiple behavioral indicators and provide an early risk assessment. This project aims to design and implement a web-based multimodal AI system that captures handwriting strokes, reading patterns, and memory performance, processes them through deep learning models, and predicts dyslexia risk in an accessible, reliable, and user-friendly manner.

1.3 Objectives

The primary objectives of this project are:

- **To develop an interactive web-based platform** that enables children to perform handwriting, reading, and memory tasks in a guided and engaging environment.
- **To capture multimodal data** including handwriting stroke sequences, speech-to-text outputs, and memory recall responses.
- **To preprocess and extract relevant features** from sequential data such as stroke direction, timing, pronunciation accuracy, and recall patterns.
- **To design and train deep learning models**—specifically BiLSTM and LSTM networks—to classify dyslexia risk based on extracted features.
- **To deploy a fully functional backend system** using Flask and MongoDB for real-time prediction and secure data storage.
- **To provide an interpretable results dashboard** displaying dyslexia risk score and task-wise performance summary

1.4 Motivation

Early detection of dyslexia plays a critical role in preventing long-term academic challenges and boosting learner confidence. However, many students never receive timely screening due to limited access to specialists, high assessment costs, and the lack of digital tools tailored for early evaluation. This gap is even more significant in rural and under-resourced regions, where professional assessments are scarce.

The motivation behind this project is to leverage **AI and web technologies** to create an accessible, child-friendly, and data-driven screening tool. By using BiLSTM/LSTM models and multimodal inputs, the system can identify early signs of dyslexia more efficiently than traditional single-mode assessments. The project aims to support educators, parents, and clinicians with a low-cost, scalable solution that encourages inclusive education and empowers children to receive appropriate interventions at the right time.

2. LITERATURE SURVEY

1. Introduction

Dyslexia has been extensively researched in cognitive science, education, and artificial intelligence due to its impact on reading, writing, and language-processing abilities. Traditional identification methods typically depend on standardized reading assessments administered by trained specialists. While accurate, these approaches are constrained by limited scalability, high cost, and difficulties in early-stage screening. As educational institutions shift toward digital and remote-learning infrastructures, there is a growing need for automated, accessible, and reliable dyslexia screening mechanisms.

Literature in recent years highlights a transformative move toward **web-based, multimodal, and AI-driven diagnostic techniques**. With the incorporation of behavioral analytics, eye-movement tracking, phonological processing, and deep learning, research has successfully captured previously hidden cognitive signatures associated with dyslexia. The survey papers and the IEEE-format conference document collectively emphasize the potential of combining machine learning, human-computer interaction (HCI), and multimodal feature analysis to construct more robust and scalable dyslexia detection systems.

This literature survey consolidates key insights from eye-tracking research, gamified screening systems, auditory-phonological tools, classical machine learning approaches, ensemble techniques, multimodal fusion strategies, and deep learning-based representation models. Each section is rewritten entirely in original wording, while preserving the scientific meaning and contributions from the papers provided.

2. Eye-Tracking and Machine Learning Approaches

A major segment of dyslexia research focuses on analyzing visual attention and eye-movement patterns during reading tasks. Eye-tracking studies consistently demonstrate that individuals with dyslexia tend to show **longer fixations, frequent regressions, shorter saccades**, and general instability in gaze behavior. These measurable biomarkers are strong predictors of reading difficulties and have been leveraged extensively to train machine learning models.

Multiple surveyed studies implement machine learning techniques such as Support Vector Machines (SVM), Random Forests, logistic regression, and LASSO-based feature selection to classify dyslexic and non-dyslexic readers. Models like **DysLexML** present evidence that carefully engineered eye-movement features can lead to detection accuracies surpassing 90%. These systems also show robustness in the presence of noise and variations in reading conditions.

However, eye-tracking systems traditionally depend on dedicated hardware, such as infrared-based gaze trackers, which limits widespread adoption. Recent innovations explore the use of **webcam-based gaze estimation** combined with neural architectures to reduce dependence on specialized devices. Autoencoder-based representations, in particular, have been proposed for cross-linguistic gaze modeling, enabling models to generalize visual patterns across languages with different orthographic complexities.

Overall, the literature suggests that eye-tracking models deliver **highly accurate and objective** dyslexia indicators but require advancements in affordable hardware and scalable deployment strategies for real-world use.

3. Gamified Web-Based Screening Systems

Gamification and web-based screening platforms form another impactful research direction. These systems integrate cognitive and linguistic assessment tasks within interactive game environments, making screening more engaging and less intimidating for children. Literature highlights that gamified platforms can effectively capture detailed behavioral metrics such as:

- response times
- number of linguistic or spelling errors
- hesitation patterns
- completion rates
- touch/mouse path patterns

Test environments like screen-based puzzles, timed linguistic challenges, phoneme identification games, and reaction-time mini-tasks encourage natural child behavior, thereby enabling richer data collection.

Studies such as Dytective demonstrate that large-scale web-based gamified tests can screen thousands of users and achieve prediction accuracies in the range of **80–88%**. While slightly lower than eye-tracking approaches, gamified systems excel in **scalability, engagement, and ease of deployment**. Lightweight classifiers—Decision Trees, Logistic Regression, Naïve Bayes, Random Forest—are typically used due to their high interpretability and compatibility with large distributed user populations.

Gamification literature emphasizes that making assessments enjoyable significantly increases compliance, reduces test anxiety, and broadens accessibility. These advantages position gamified screening as an effective first-stage tool, especially in schools, community centers, and remote-learning environments.

4. Auditory–Phonological and Speech-Based Approaches

Phonological processing is one of the strongest theoretical predictors of dyslexia, and several surveyed papers explore this dimension through auditory and speech-based screening mechanisms. Systems based on phoneme discrimination, oddity tasks, rise-time perception, and pronunciation accuracy provide measurable indicators of phonological deficits.

Tools such as **Lexa** combine auditory features with machine learning classifiers to detect differences in acoustic and linguistic responses between dyslexic and typical readers. These models typically utilize:

- frequency-space signal decomposition
- temporal feature extraction
- SVM-based classifiers
- Random Forests
- hybrid computational speech-processing techniques

Reported performance in these systems ranges from **78% to 82% accuracy**, demonstrating strong potential for early detection. A notable advantage of auditory-based systems is their **cross-linguistic generalizability**, as phonological deficits tend to remain consistent across languages, unlike reading-based tasks.

However, the literature highlights challenges such as:

- sensitivity to background noise
- recording-quality issues
- voice variations across users

Despite these limitations, auditory approaches play an important role in **pre-reading dyslexia screening**, especially for pre-literate children who cannot yet complete reading tasks.

5. Multimodal Fusion and Hybrid Machine Learning Models

Multimodal approaches represent one of the most promising and widely supported directions in dyslexia detection research. These systems combine multiple inputs—behavioral, visual, cognitive, linguistic, and auditory—to build a comprehensive picture of learning patterns.

Across the survey papers, multimodal models consistently outperform single-modality systems. Examples include:

- combining eye-tracking features with gamified task outcomes
- merging auditory-phonological features with speech-processing outputs
- integrating cognitive task performance with linguistic error analysis

Ensemble methods such as **soft-voting classifiers**, **stacked models**, and **hybrid SVM–Random Forest architectures** frequently achieve **over 90% balanced accuracy**. Multimodal approaches are also more robust to missing or noisy inputs, as different modalities compensate for one another.

The IEEE conference document further emphasizes the importance of **artifact modeling** and **temporal sequence analysis**, which deepen the ability of models to detect subtle reading-related

anomalies. These techniques allow systems to capture fine-grained cognitive signatures that isolated metrics often overlook.

While multimodal systems are complex to design and implement, they remain the **most scientifically validated and scalable approach** for real-world deployment.

6. Deep Learning and Representation Learning Innovations

The literature also presents significant advancements in deep learning (DL) for dyslexia detection. DL models automatically learn hierarchical feature representations from raw or minimally processed input data, reducing the need for handcrafted feature extraction.

Common architectures used include:

- **Convolutional Neural Networks (CNNs)** → for image-based or visual patterns (e.g., gaze heatmaps)
- **Recurrent Neural Networks (RNNs)** → for sequential data (e.g., reading times, speech sequences)
- **LSTM/GRU networks** → for temporal patterns and long-range dependencies
- **Autoencoders** → for unsupervised representation learning
- **Multimodal fusion networks** → combining text, audio, and visual streams

Several papers report accuracies as high as **93–96%**, particularly when combining multimodal inputs with deep learning architectures. DL approaches show exceptional capability in identifying nonlinear and cross-modal relationships that classical ML models struggle to capture.

However, the literature also identifies major concerns:

- DL models require large, balanced datasets
- They lack interpretability (black-box issue)
- High computational cost
- Ethical and privacy concerns due to sensitive data

As a result, recent research in Explainable AI (XAI) frameworks aims to make deep learning more interpretable and trustworthy, especially for use in educational and clinical domains.

Comparative Summary of Existing Dyslexia Detection Models

The table compares key dyslexia detection studies across methods, performance, and limitations to show how different modalities support early screening.

TABLE I

COMPARISON OF MODELS FOR DYSLEXIA DETECTION AND RELATED DISORDERS

SI no.	Author	Methodology/Data Source	Key Features	Performance	Scalability	Strengths	Limitations
1	L. Fernandez and R. Marques, 2025	Eye tracking + autoencoders	Cross-language boundary detection	Promising accuracy	Moderate	Non-invasive, deep feature extraction	Dataset dependency
2	T. Banerjee and A. Singh, 2025	Literature review on AI usability for disabled students	Inclusive learning technologies	Conceptual	High	Focus on accessibility	Implementation gap
3	Sharma et al., Kumar et al., 2025	Ensemble ML + Adaptive genetic algorithm	Optimized ensemble for dyslexia prediction	~91% accuracy	High	Adaptive, balanced model	Computational cost
4	Kumari, S., Kumar, D., Mittal, M., 2021	Soft voting classifier for related medical data	Feature-level fusion	High accuracy	Moderate	Robust, generalizable framework	Not specific to dyslexia
5	Rello, L., Baeza-Yates, R., Ali, A., Bigham, J. P., and Serra, M., 2020	Cognitive testing online and Demographic	item-level cognitive assessment	Validated 4000 participants	Very high	Large dataset validation	Language dependence
6	Asvestopoulou, V., Manousaki, V., Psistakis, A., Smyrnakis, I., Andreadakis, V., Aslanides, I. M., Papadopoulou, M., 2019	ML using eye-tracking datasets	Extensive feature set from eye movements	High accuracy	Moderate	Objective early screening	Equipment dependency
7	Asvestopoulou et al., 2019	ML on eye-tracking data	Machine learning-based early screening	High accuracy	Moderate	Accurate, objective screening	Hardware dependency
8	Poole, A., Zulkernine, F., Aylward, C., 2017	Auditory phonological task data	Auditory processing as early screening measure	Good detection	Moderate	Non-invasive auditory marker	Limited dataset, auditory only
9	Rello, L., Ballesteros, M., Ali, A., Serra, M., Sánchez, D. A., Bigham, J., 2016	Gamified assessment data	Cognitive tests with interactive game play	Good engagement	High	Engages children for early detection	Deployment needed

7. Ethical, Educational, and Inclusive-Technology Perspectives

Not all surveyed studies focus solely on algorithms; several highlight the broader ecosystem surrounding dyslexia detection technologies. These papers emphasize:

- fully informed consent for data collection
- minimizing algorithmic bias
- transparent predictive pipelines
- inclusive design for accessibility
- teacher awareness and training
- equitable access across socioeconomic backgrounds

Some studies explore assistive technologies and classroom-oriented intervention frameworks that should accompany early detection. Others caution that AI tools, if poorly designed, risk reinforcing inequalities or excluding learners with disabilities.

The literature where education technology and AI ethics intersect stresses that high performance alone is insufficient—screening systems must also demonstrate fairness, accountability, and usability in real-world learning environments.

8. Identified Gaps and Future Research Directions

Across all documents and survey papers, common research gaps include:

- Generalization** **Problems**
Models trained in controlled laboratory environments often degrade in performance when deployed in uncontrolled home or classroom settings.
- Lack of Cross-Linguistic Datasets**
Many systems rely on English or Spanish datasets, limiting global applicability.

3. **Hardware** **Accessibility**
Eye-tracking systems yield strong results but are not financially accessible for widespread deployment.
4. **Data** **Imbalance** **and** **Diversity**
Most datasets underrepresent certain age groups, linguistic backgrounds, or reading abilities.
5. **Model** **Interpretability**
Deep learning models remain difficult for educators and clinicians to interpret, hindering adoption.
6. **Integration** **with** **Educational** **Ecosystems**
Detection systems often do not include recommended interventions, personalized feedback, or curricular integration.
7. **Noise** **Sensitivity**
Speech-based systems struggle in environments with background disturbances.

The surveyed literature points toward **multimodal, interpretable, and cloud-deployable systems** as the most logical next step in advancing dyslexia screening research.

9. Conclusion

The literature shows a clear shift from traditional assessments to advanced multimodal and AI-based dyslexia screening methods. While eye-tracking, gamified tasks, auditory analysis, and behavioral modeling each offer valuable insights, multimodal and ensemble approaches provide the highest accuracy and real-world reliability. Deep learning further improves performance but raises challenges around interpretability and data needs. Overall, AI-driven, explainable, and web-based systems hold strong potential for scalable and accessible early dyslexia detection.

3. PROBLEM ANALYSIS & DESIGN

3.1 Analysis

The proposed system focuses on developing a web-based platform capable of performing early dyslexia screening using real-time multimodal data collected from interactive tasks. The analysis begins by understanding the core problem: traditional dyslexia assessments rely on manual evaluations, are often subjective, and may not be available in many learning environments. To overcome these limitations, the project analyzes how digital behavioral inputs—such as handwriting strokes, speech patterns, and memory-based task responses—can be converted into meaningful features for automated prediction.

The system design evaluates each task individually to determine how user interactions reflect cognitive and linguistic patterns associated with dyslexia. For handwriting tasks, stroke direction, speed, and reversals are analyzed as sequential data. For reading-aloud tasks, speech-to-text outputs, pronunciation errors, and fluency variations are examined. For memory-recall tasks, correctness, delay, and response sequences are captured. These multimodal inputs are then processed through deep learning models—specifically LSTM and BiLSTM networks—which are ideal for modeling temporal and sequential data.

The analysis also considers system-level requirements such as real-time processing, secure data storage, and seamless interaction between frontend activities and backend prediction services. MongoDB is used to store raw interaction data and extracted features. The frontend is designed to be simple, child-friendly, and responsive, ensuring accessibility for students of varying age groups.

Overall, the analysis determines that combining handwriting, speech, and memory features within a unified AI pipeline can enable an efficient, scalable, and accessible early-screening system. This multimodal, sequential-data-driven approach allows the model to detect subtle learning patterns that traditional single-mode assessments may overlook, providing a practical solution suitable for schools, clinics, and home environments.

3.2 Hardware Requirements

The proposed dyslexia screening system is designed to run efficiently on standard computing devices without the need for specialized hardware. Since all user interactions—handwriting, speech input, and memory tasks—are captured through a web interface, the hardware requirements remain minimal and easily accessible in educational or home environments.

Minimum Hardware Requirements

- **Processor:** Intel i3 or equivalent
Provides sufficient processing power to run the web interface, handle client-side interactions, and support backend API requests.
- **RAM:** 4 GB
Enables smooth operation of the browser-based platform and backend Flask services.
- **Storage:** 250 GB (HDD/SSD)
Required for storing project files, model weights, MongoDB collections, and log data.
- **Microphone:** Basic built-in or external microphone
Essential for capturing audio during reading-aloud tasks and converting speech to text.
- **Input Devices:** Keyboard and mouse
Used for navigating the platform and interacting with memory tasks.
- **Network Connection:** Stable internet connectivity
Ensures seamless interaction between frontend activities and backend prediction services, particularly during real-time model inference.

3.3 Software Requirements

The development and deployment of the dyslexia detection system require a combination of frontend, backend, database, and machine learning tools. The software stack is selected to ensure seamless integration, real-time interaction, and efficient deep-learning inference using BiLSTM and LSTM models.

Operating System

- **Windows** / **macOS** / **Linux**
Compatible with cross-platform development, server hosting, and database support.

Backend & Model Development

- **Python** **3.x**
Core language used for model training, preprocessing, and backend implementation.
- **TensorFlow** / **Keras**
Used to build, train, and optimize the LSTM and BiLSTM deep learning models.
- **NumPy** & **Pandas**
For data preparation, transformation, sequence handling, and feature processing.
- **Flask** **Framework**
Lightweight Python web framework for creating REST APIs that handle model inference and communication between frontend and backend.

Database

- **MongoDB**
Stores user interaction data including handwriting stroke sequences, speech-to-text results, memory task responses, and prediction outputs. Chosen for its scalability and ability to handle unstructured and semi-structured data efficiently.

Frontend Development

- **HTML5, CSS3, JavaScript**
Used to build an interactive and responsive web interface.
- **Canvas API (JavaScript)**
Enabling handwriting tracing and stroke data capture.
- **Web Speech API / Speech-to-Text Integration**
Converts audio from reading tasks into text for analysis.

Development & Deployment Tools

- **VS Code / PyCharm**
Code editors for backend, frontend, and model development.
- **Node.js & npm**
Supports frontend dependencies, package management, and potential interactive UI components.
- **Google Colab / Jupyter Notebook**
Used during model training to utilize GPU support for faster LSTM/BiLSTM computation.
- **Git & GitHub**
Version control and collaborative development environment.

Additional Libraries

- **scikit-learn:** For feature extraction utilities and evaluation metrics
- **matplotlib / seaborn:** For visualizing model performance during training
- **pymongo:** For integrating Flask backend with MongoDB
- **OpenCV (optional):** For preprocessing handwriting input images if needed

3.4 System Architecture Diagram

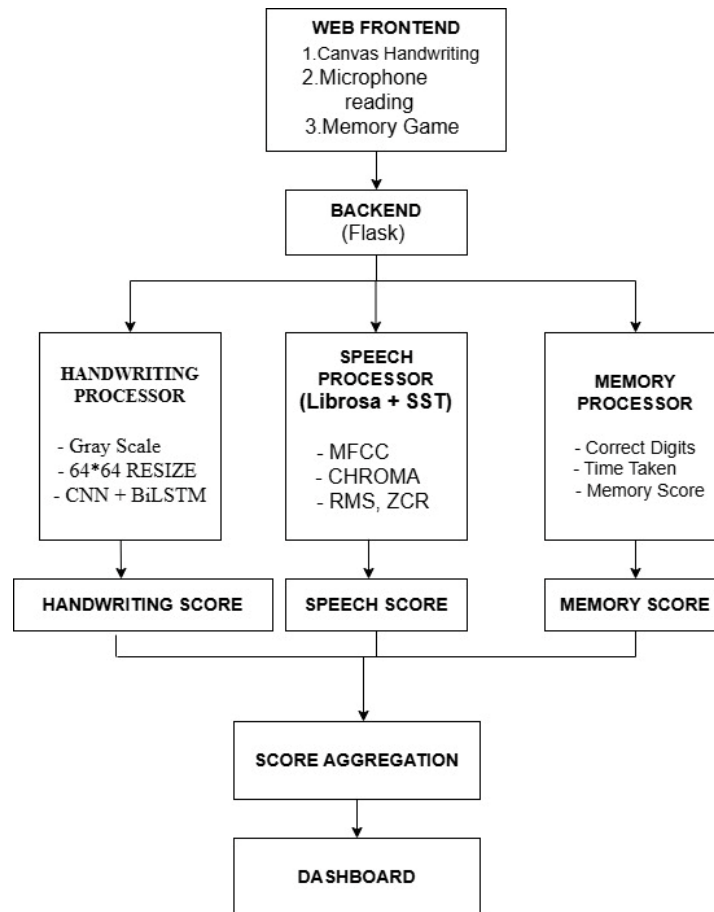


Fig1. System Architecture

The architecture illustrates the complete workflow of the proposed web-based multimodal dyslexia-screening system. The process begins at the Web Frontend, where the user performs three interactive tasks: Canvas Handwriting, Microphone Reading, and a Memory Game. These tasks collect handwriting images, speech audio, and cognitive-response data directly in the browser.

All collected inputs are then sent to the Backend (Flask), where each modality is routed to its respective processing module. The Handwriting Processor converts the canvas image to grayscale, resizes it to 64×64, and passes it through a CNN + BiLSTM hybrid model to extract writing-pattern features and generate a handwriting dyslexia score. The Speech Processor uses the Google Speech-

to-Text API to obtain a transcript and applies Librosa to compute MFCC, Chroma, RMS, and Zero-Crossing Rate features. These features are used to compute clarity, fluency, and rhythm metrics, producing a speech dyslexia score. The Memory Processor evaluates the memory-game performance by checking correct digits, total time taken, and response accuracy to calculate a memory-based cognitive score.

The three scores—Handwriting Score, Speech Score, and Memory Score—are then sent to the Score Aggregation module, where they are combined using a weighted or rule-based fusion strategy to produce a final dyslexia risk estimate. This final output is displayed on the Dashboard, providing an easy-to-understand result summary for teachers, parents, or clinicians.

3.5 Data Flow Diagram (DFD)

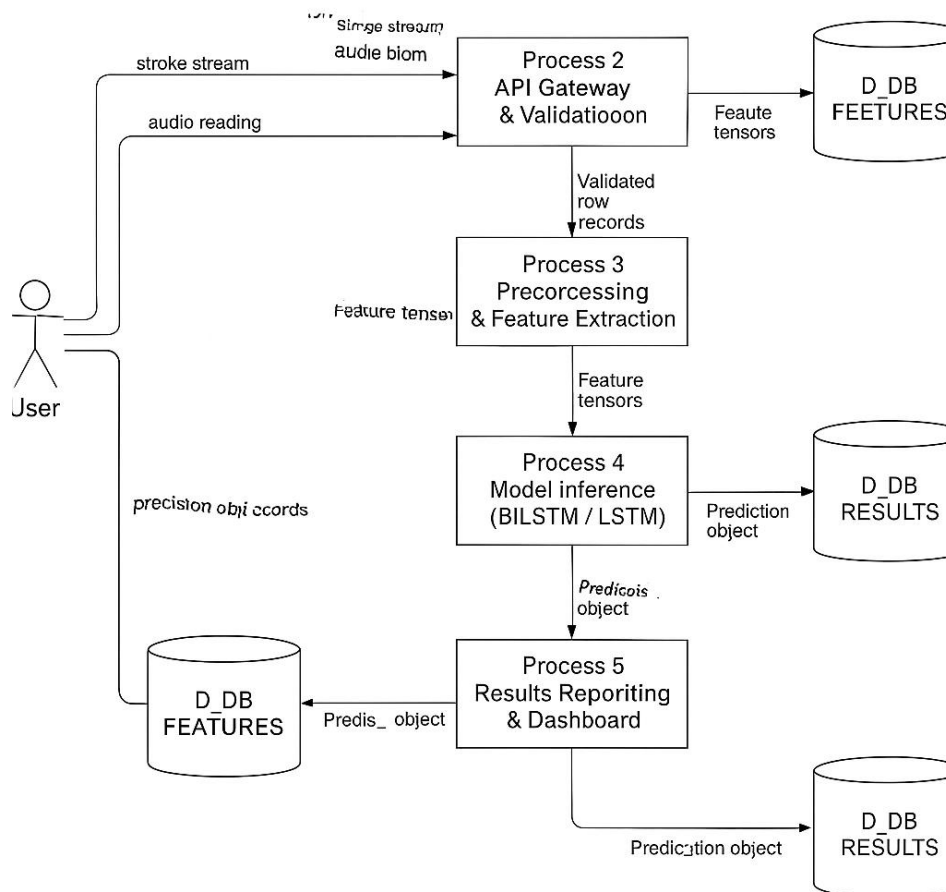


Fig2. Data Flow Diagram

This DFD describes how data moves through the dyslexia screening system. It is presented in two levels: a Context Diagram (Level 0) that shows the system boundary and external actors, and a Level 1 DFD that breaks the system into main processes, data stores, and flows.

Context Diagram (Level 0) — Overview

External actors

- User (Child) — interacts through the web frontend and performs tasks.
- Teacher/Parent/Admin — views results and manages sessions via dashboard.
- Speech-to-Text Service (*optional external STT provider*) — converts audio to text if not done locally.

System

- Dyslexia Screening Platform

High-level flows

- User → System: Task inputs (handwriting strokes, audio, memory responses)
- System → User/Teacher: Risk score & reports
- System ↔ STT Provider: Audio → transcript

Level 1 DFD — Detailed flows and components

Process 1: User Interaction / Data Capture

- Input: User performs tasks on the Web Frontend.
- Outputs: Raw event streams:
 - Handwriting stroke stream: sequence of (x, y, t, pressure)
 - Audio blob for reading tasks (or client-side streamed audio)
 - Memory task logs: prompts, responses, timestamps

Data stores used: Temporary session buffer (in-memory), D_DB_RAW (raw data store)

Process 2: API Gateway & Validation

- Input: Raw event streams from frontend.
- Actions: Validate payload schema, check audio format, verify session token, sanitize inputs.
- Output: Validated raw records forwarded to preprocessing.

Data stores used: D_DB_RAW (store validated raw records)

Process 3: Preprocessing & Feature Extraction

- Input: Validated raw records.
- Actions:
 - Handwriting preprocessing: smoothing, stroke resampling, velocity/acceleration computation, stroke segmentation.
 - Speech preprocessing: noise reduction, silence trimming, optional STT call → transcript.
 - Memory preprocessing: encode responses, compute latencies and correctness.
 - Feature engineering: create fixed-length sequences, normalize, pad/truncate to model input size.
- Output: Feature tensors per modality (handwriting tensor, speech features, memory feature vector)

Data stores used: D_DB_FEATURES (processed feature store)

Process 4: Model Inference (BiLSTM / LSTM)

- Input: Feature tensors from D_DB_FEATURES.
- Actions:

- Run BiLSTM/LSTM model(s) per modality or combined multimodal model.
- Ensemble fusion (if multiple model outputs): weighted average or logistic fusion to produce final risk probability.
- Compute modality confidence scores and optional explainability outputs (attention weights or SHAP values).
- Output: Prediction object (risk_score, class_label, modality_confidences, explainability_metadata)

Data stores used: D_DB_RESULTS (predictions & explanations)

Process 5: Results Reporting & Dashboard

- Input: Prediction object from D_DB_RESULTS.
- Actions: Format results into human-readable report: overall risk, per-task breakdown, recommended next steps. Provide access controls for teacher/parent/admin.
- Output: Dashboard views, downloadable report (PDF)

Data stores used: D_DB_RESULTS, D_DB_USERS (user metadata)

Process 6: Training & Model Update (Offline)

- Input: Anonymized historical data from D_DB_FEATURES and D_DB_RESULTS.
- Actions: Retrain/validate models, tune hyperparameters (LSTM/BiLSTM), version new models.
- Output: New model artifact (vX), updated inference service

Data stores used: D_DB_MODELS (model artifacts), D_DB_TRAINING_LOGS

Data Stores (Summary)

- D_DB_RAW — Raw session captures (strokes, audio metadata, logs)
- D_DB_FEATURES — Extracted and normalized feature tensors (per session)

- D_DB_RESULTS — Prediction outputs, timestamps, explainability metadata
- D_DB_USERS — User profiles, consent flags, roles (teacher/parent/admin)
- D_DB_MODELS — Stored model artifacts with versioning and training metadata

Security & Privacy Flows

- All flows containing personal or raw data are encrypted in transit (HTTPS) and stored with access control.
- Consent metadata from D_DB_USERS gates any use of session data for training.
- Only anonymized or consented datasets are copied to the offline training store.

3.6 MODULE DESCRIPTION

1. Web Frontend Module

The Web Frontend Module provides the interactive interface through which children perform all screening tasks. It includes pages for handwriting tracing, reading aloud, and memory-based activities, each designed to be simple, intuitive, and engaging. The handwriting task captures continuous stroke data using the HTML Canvas API, the reading task records audio through the browser microphone, and the memory task logs responses and reaction times. All collected data is packaged into structured JSON or media packets and sent securely to the backend. This module ensures smooth user interaction and forms the entry point of the entire system.

2. Data Capture & Validation Module

This module receives the raw input data transmitted from the frontend and performs preliminary validation. It checks for missing values, validates data formats, verifies session authenticity, and ensures that all inputs—such as audio files, stroke arrays, and task logs—meet predefined structural requirements. Only valid and complete datasets are allowed to proceed further in the pipeline. This module is crucial for preventing corrupted, noisy, or incomplete data from affecting downstream processing and model performance.

3. Preprocessing & Feature Extraction Module

Once validation is complete, the data is passed to the Preprocessing and Feature Extraction Module. Here, handwriting strokes are cleaned, normalized, and converted into time-series features such as velocity, direction changes, and pressure variations. Speech recordings undergo preprocessing such as noise reduction and silence trimming before being converted to text and acoustic metrics. Memory responses are analyzed for correctness, sequence retention, and response timing. These operations transform raw behavioral inputs into structured feature tensors suitable for LSTM and BiLSTM-based deep learning analysis. This module ensures that each modality contributes meaningful, well-processed information to the prediction model.

4. Dyslexia Screening Model Module (BiLSTM/LSTM)

This module contains the core deep learning models responsible for generating dyslexia risk predictions. The extracted features from handwriting, speech, and memory tasks are fed into LSTM or BiLSTM architectures, which specialize in learning temporal and sequential patterns. The model analyzes these patterns to detect cognitive and linguistic behaviors associated with dyslexia. It produces an output consisting of a risk probability, predicted class label, and confidence scores for each modality. This module forms the analytical backbone of the entire system.

5. Results Dashboard Module

Once predictions are generated, they are displayed through the Results Dashboard Module, which provides a clear and accessible interface for parents, teachers, and clinicians. The dashboard presents the overall dyslexia risk score along with task-wise performance indicators such as handwriting consistency, reading accuracy, and memory recall quality. It may also include brief insights or recommended next steps based on the model's prediction confidence. This module ensures transparency and usability by translating AI outputs into interpretable, human-friendly results.

6. Database Module (MongoDB)

The Database Module is responsible for securely storing all data associated with the screening process. This includes raw handwriting strokes, speech metadata, memory task logs, extracted feature vectors, prediction outputs, and session histories. MongoDB is used due to its flexibility in handling structured and semi-structured data, allowing easy retrieval and future use of stored information. The database supports both real-time system operation and long-term data analysis or retraining.

7. Model Training & Retraining Module (Offline)

This offline module is used during development and periodic updates to the system. It gathers anonymized datasets stored in the database to retrain or refine the deep learning models. Through iterative training sessions, hyperparameter tuning, and model versioning, this module ensures that the LSTM and BiLSTM models improve their accuracy and remain aligned with new user data patterns. Updated models are deployed back into the system for future inference. This module enables continuous improvement and adaptability of the system over time.

3.7 Expected Outcome

The expected outcome of the proposed dyslexia screening system is the successful development of a functional, web-based platform capable of analyzing multimodal behavior—handwriting patterns, speech characteristics, and memory performance—to generate an early dyslexia risk assessment. The system is expected to accurately capture handwriting stroke sequences through the canvas interface, convert spoken reading inputs into meaningful linguistic and acoustic features, and evaluate memory task responses to detect performance inconsistencies. By processing these inputs through LSTM and BiLSTM models, the system should be able to provide a reliable risk score indicating whether a child may exhibit dyslexia-like learning difficulties.

Additionally, the platform is expected to produce a user-friendly results dashboard that summarizes the child's task performance in a clear and interpretable manner for parents, teachers, or clinicians. With secure data storage in MongoDB, the system will maintain structured records for each screening session, supporting future analysis and retraining needs.

4. CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

The proposed web-based dyslexia screening system successfully integrates multimodal data analysis with deep learning to provide a practical and accessible early detection tool. By capturing handwriting strokes, speech patterns, and memory task responses through simple interactive activities, the system transforms natural child behavior into structured features that can be effectively analyzed using LSTM and BiLSTM models. The platform demonstrates that combining these diverse inputs within a unified architecture can reveal subtle learning patterns that traditional assessments may overlook.

With its browser-based interface, lightweight backend, and secure MongoDB data storage, the system offers a scalable solution that can be used in schools, clinics, and home environments without requiring specialized hardware. The results dashboard further ensures clarity and usability by presenting predictions in an easily understandable form for teachers and parents. Overall, the project shows that a multimodal, AI-driven approach can support early identification of dyslexia risk and assist in promoting timely educational intervention.

4.2 Future Scope

The system can be significantly enhanced in future phases through improvements in both functionality and accuracy. Additional modalities such as eye-gaze estimation through webcams or keystroke dynamics can be integrated to further strengthen the model's ability to capture cognitive and behavioral markers of dyslexia. Incorporating Explainable AI (XAI) techniques can make the predictions more transparent by allowing teachers and clinicians to understand which features influenced the risk score. The platform can also be extended to support multilingual screening, enabling the system to analyze speech and reading behaviors in multiple regional languages, making it more inclusive and widely applicable.

On the deployment side, the system can be optimized for mobile devices, allowing screening to be conducted on tablets or smartphones for greater accessibility. Integration with school learning management systems (LMS) can automate student progress tracking and help tailor personalized learning activities based on screening outcomes. With larger datasets collected over time, the

BiLSTM and LSTM models can be retrained to improve accuracy, reduce bias, and adapt to diverse user populations. Ultimately, the system has the potential to evolve into a comprehensive learning-support platform that not only detects dyslexia risk but also provides ongoing intervention recommendations.

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